POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

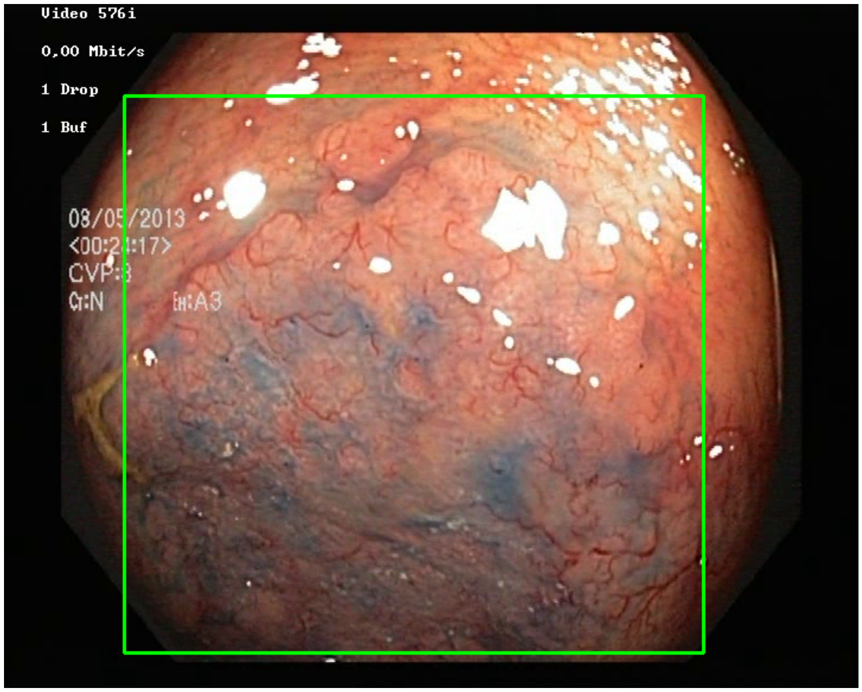
Video ID:	a6340694-7315-4e1b-b8a5-c048df133d79
Total Frames:	686
Frame Rate:	25.0 fps (assumed)
Duration:	27.4 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	293	✓
RT-DETR Detections	89	✓
MedSAM2 Detections	331	✓
Consensus (3-Model Agreement)	1	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	1	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 210 frames)



Feature	Value	Interpretation
Frame Index	37	Frame 37 of 686
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.194	Color-based vascularization
Relative Radius	10.0%	Polyp size as % of frame diagonal
Texture Score	0.184	Surface morphology
Vessel Visibility	0.924	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	69.9%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:
A total of 1 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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